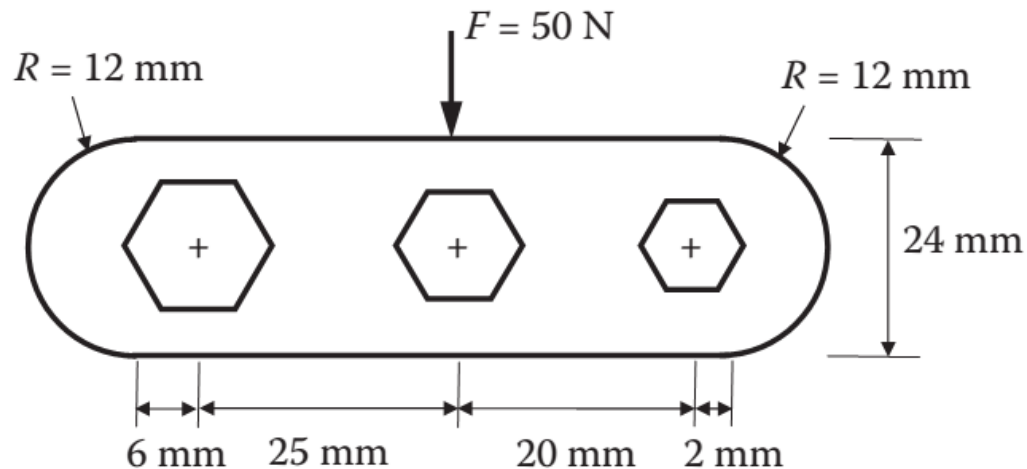


Stresses and deflection analysis on frames

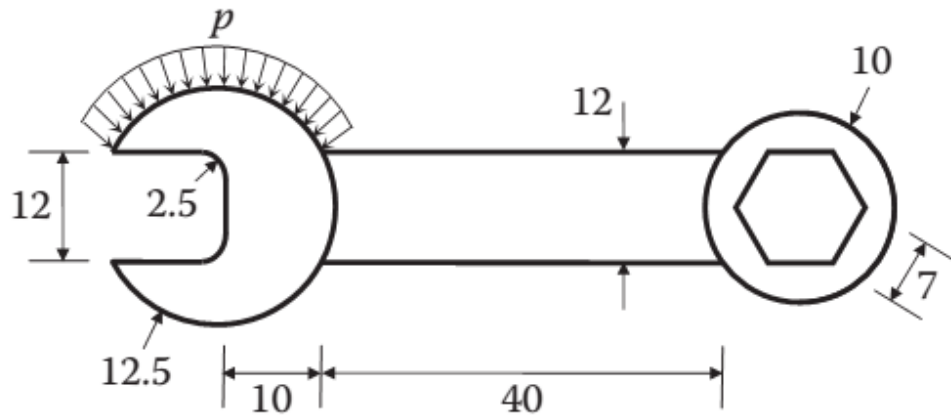
A bicycle wrench shown below is made of stainless steel with a Young's modulus of **193 GPa** and a Poisson's ratio of **0.27**. The wrench is **2 mm thick**. The side lengths of the hexagons from left to right are **9, 7, and 5 mm**, respectively. The smallest hexagon is fixed on all sides. A **50N** force is applied on the top of the wrench.

Using *ANSYS Workbench*, determine:

1. The location and magnitude of the maximum deformation
2. The maximum von Mises stress under the given load and boundary conditions.



A combination wrench is a convenient tool that is used to apply torque to loosen or tighten a fastener. The wrench shown below is made of stainless steel and has a thickness of 3 mm. Using ANSYS Workbench, determine the maximum deformation and the distribution of von Mises stresses under the given distributed load and boundary conditions.



Material: Stainless steel

$E = 193 \text{ GPa}$

$\nu = 0.27$

Boundary conditions:

Hexagon on the right end is fixed on all sides.

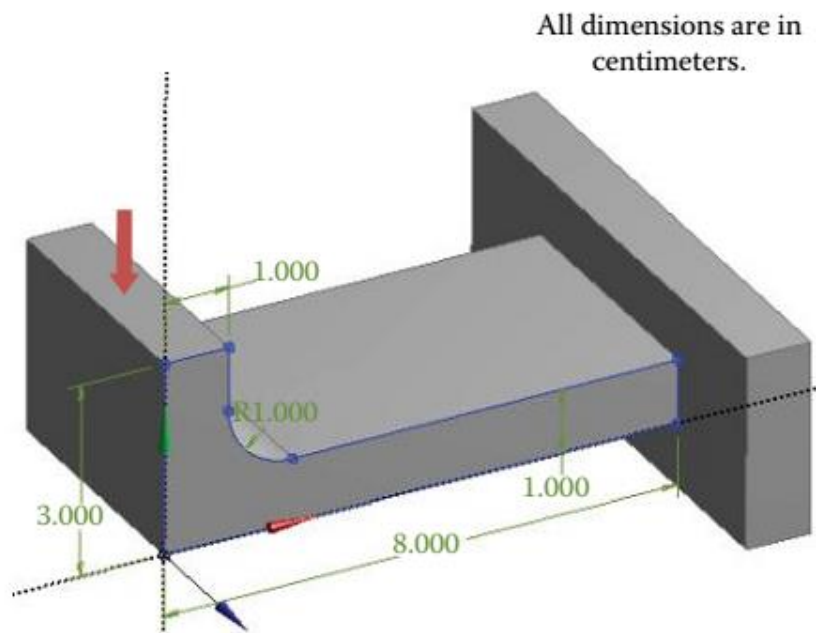
Pressure $p = 2 \text{ MPa}$

(All units are in millimeters.)

The following figure shows an L-shaped bracket made of structural steel with Young's modulus $E = 200 \text{ GPa}$ and Poisson's ratio $\nu = 0.3$. The **4 cm thick** L-shape with a cross-section profile shown in the figure is rigidly connected to a rectangular base of size $3 \text{ cm} \times 8 \text{ cm} \times 1 \text{ cm}$. Suppose that the bracket is fixed on the right end at the base. A uniform pressure of 100 N/cm^2 is applied on the protruding surface of the L-shape as shown below.

Using Ansys WB, it is required to:

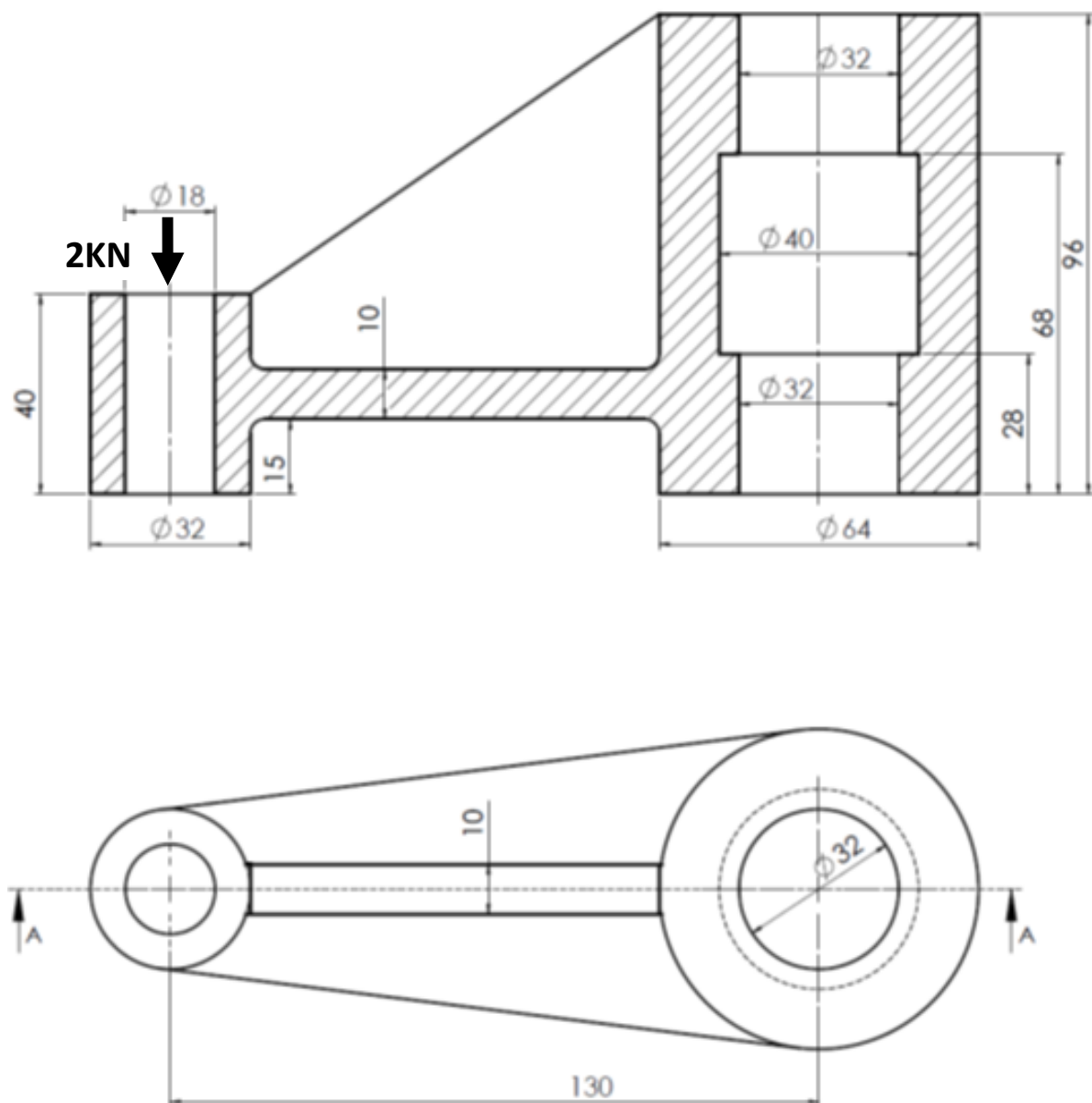
- 1-Determine the **deformation** and **stresses** in the bracket.
- 2-Make parametric analysis by changing the length (8cm).



The following figure shows a bracket made of structural steel with Young's modulus $E = 200 \text{ GPa}$ and Poisson's ratio $\nu = 0.3$. The lower surface of the bracket at $\phi 64$ is considered completely fixed and the bracket is loaded with 2KN acting on the area between $\phi 32$ and $\phi 18$ due to a hinge pin mounted in hole $\phi 18$.

Using Ansys WB, it is required to:

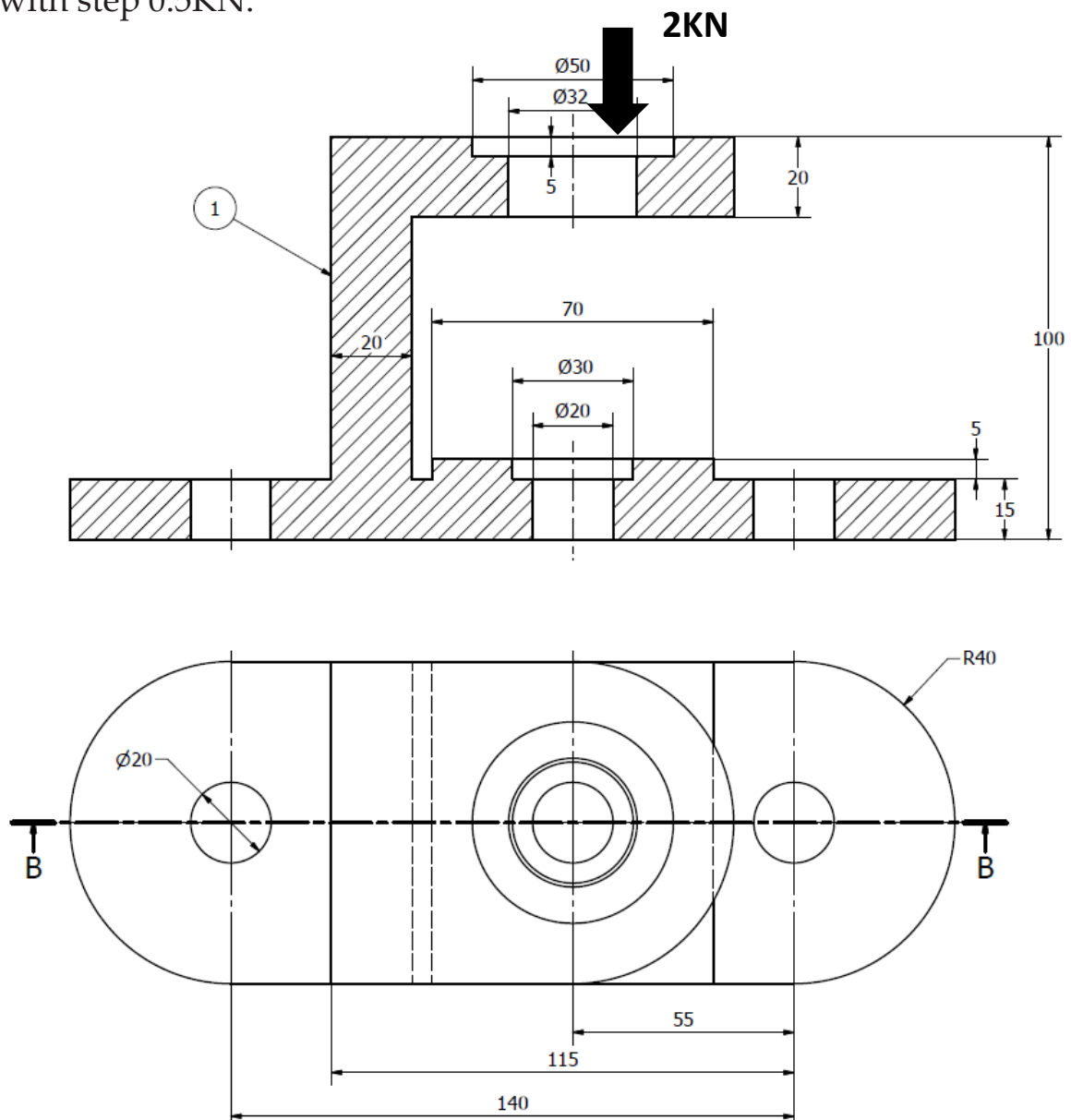
- 1-Determine the **deformation** and **stresses** in the bracket.
- 2-Make parametric analysis by changing the length (130mm).



The following figure shows a bracket made of structural steel with Young's modulus $E = 200 \text{ GPa}$ and Poisson's ratio $\nu = 0.3$. The lower surface of the bracket is considered completely fixed and the bracket is loaded with 2KN acting on the area between $\phi 50$ and $\phi 32$.

Using Ansys WB, it is required to:

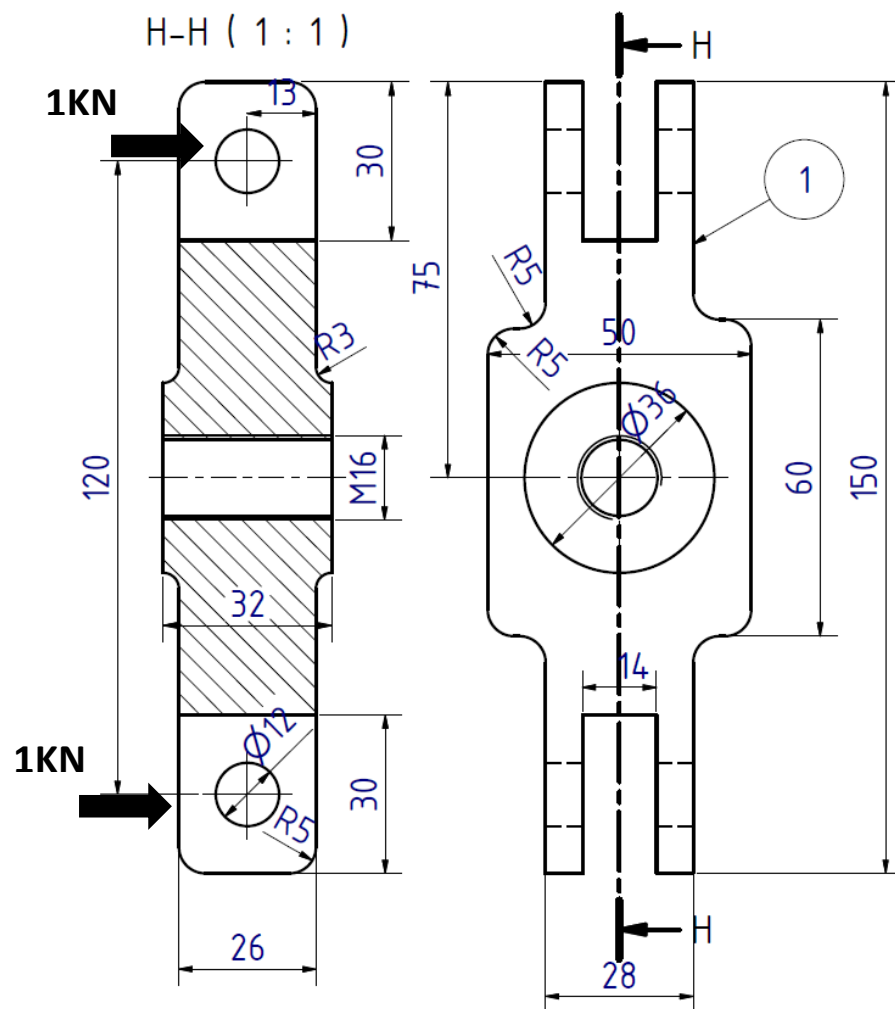
- 1-Determine the **deformation** and **stresses** in the bracket.
- 2-Make parametric analysis by changing load from 2KN to 4KN with step 0.5KN .



The following figure shows a bracket made of structural steel with Young's modulus $E = 200 \text{ GPa}$ and Poisson's ratio $\nu = 0.3$. The clamp is considered locked by a power screw of **M16** in the middle. The clamp is loaded with **1kN** acting on both sides on hole $\phi 12$ due to the presence of hinge pins.

Using Ansys WB, it is required to:

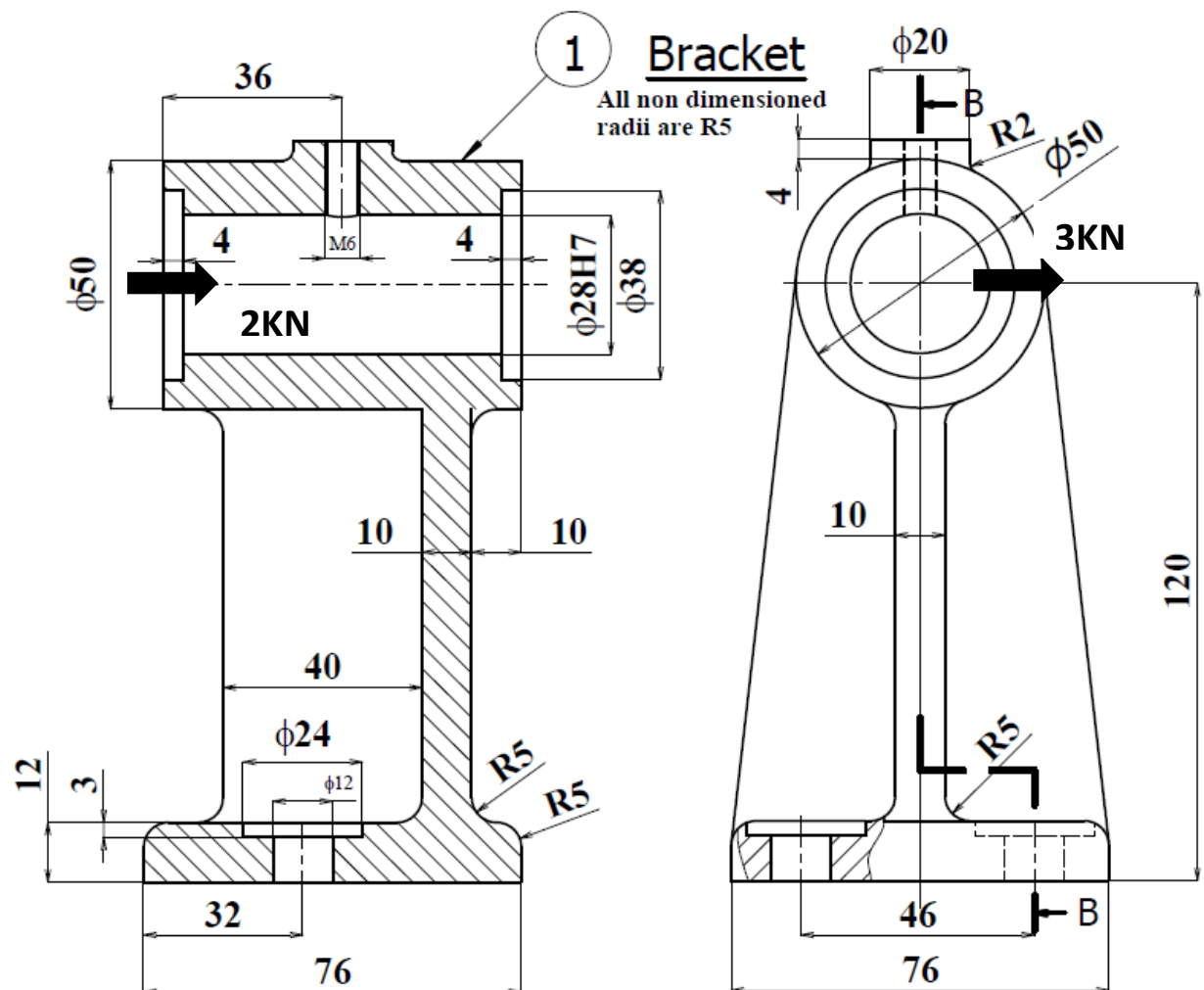
- 1-Determine the **deformation** and **stresses** in the bracket.
- 2-Make parametric analysis by changing the load from 1kN to 2kN with step 0.25kN.



The following figure shows a bracket made of structural steel with Young's modulus $E = 200 \text{ GPa}$ and Poisson's ratio $\nu = 0.3$. The bracket is fixed by 4 bolts on the base. The bracket is loaded with an axial load of **2kN** acting between $\phi 38$ and $\phi 28$, and radial load of **3kN** acting on the normal direction on hole $\phi 28$.

Using Ansys WB, it is required to:

- 1-Determine the **deformation** and **stresses** in the bracket.
- 2-Make parametric analysis by changing the axial load from 2kN to 4kN with step 0.5kN.



The following figure shows a bracket made of structural steel with Young's modulus $E = 200 \text{ GPa}$ and Poisson's ratio $\nu = 0.3$. The lower surface of the bracket is considered completely fixed and the bracket is loaded with 2KN acting on hole $\phi 12$.

Using Ansys WB, it is required to:

- 1-Determine the **deformation** and **stresses** in the bracket.
- 2-Make parametric analysis by changing load from 2KN to 4KN with step 0.5KN .

